

in India. The high quality and quantity of products and services offered by these companies have helped to transform the Indian industrial automation market.

Introduction of automation in the Indian economy has resulted in higher-quality products, faster and more efficient production (thereby reducing waste) and accidents minimisation on the production floors.

## Market drivers

With rising cost of production in China and strengthening of Yuan against dollar, investors are looking at alternate manufacturing destinations. India is aiming to become the preferred manufacturing destination for global players. However, for that, it needs to focus on quality production (zero defect) and environment-friendly manufacturing (zero effect). Both are possible only with a proper understanding and application of industrial automation.

The Make in India initiative is also expected to play an important role in boosting adoption of factory automation solutions in the country. With major investments expected in the Indian manufacturing sector, the demand for factory automation solutions in the country is anticipated to rise over the next five years.

According to a TechSci Research report, factory automation solutions are seeing a growing demand from automotive hubs across India. Other sectors driving the demand include oil and gas, discrete manufacturing and energy management.

Process automation is currently under some threat because fresh investments in commodities like steel, cement and fertilisers are drying up due to global over-capacity. So, demand from those areas is expected to flatten a little. Sectors with good prospects are pharmaceuticals, food processing, engineering components, defence manufacturing and, of course, infrastructure.

## INDUSTRIAL CONTROL AND FACTORY AUTOMATION MARKET

### By Solution:

- Supervisory control and data acquisition (SCADA)
- Programmable logic controller (PLC)
- Distributed control system (DCS)
- Manufacturing execution system
- Product lifecycle management
- Functional safety
- Plant asset management
- Cyber security

### By Component:

- Industrial robots
- Control valves
- Control devices
- Field instruments
- Industrial PC
- Human-machine interface (HMI)
- Sensor
- Industrial 3D printing
- Machine vision

### By application sector/industry:

#### 1. Process

- Oil and gas
- Chemicals
- Paper and pulp
- Pharmaceuticals
- Mining and metals
- Food and beverages
- Power
- Others

#### 2. Discrete

- Automotive
- Machine manufacturing
- Semiconductor and electronics
- Aerospace and defence
- Medical devices
- Others

### Technologies enabling industrial control and factory automation in manufacturing:

- Artificial intelligence (AI)
- Big data analytics
- Industrial
- Augmented reality (AR) and virtual reality (VR)
- Industrial Internet of Things (IIoT)
- Condition monitoring

Source: Reportlinker

Automation industry offers services for other manufacturing companies in different industrial verticals. Proximity to these industrial verticals becomes a major contributor to its growth.

In India, the western region has the highest concentration of manufacturing plants, mostly in the states of Gujarat and Maharashtra (37 per cent of establishments) followed by the southern states of Karnataka, Tamil Nadu and Andhra Pradesh (27 per cent of the establishments). Every zone has a belt where automation activities have flourished with profound concentration of manufacturing activities.

TechSci reports that western and northern regions are major markets for factory automation in India due to continuing industrial growth and presence of special economic zones and various automotive hubs in these regions.

## Market challenges

Automation is not a phenomenon that happens overnight—it takes years for

the economy to adjust and evolve with the growing technology. However, evolution of automation from assistance to full adoption is likely to take several decades depending on several factors listed below.

### Skilled manpower.

The demand and supply of skilled and unskilled workers directly decide which activities will be automated. Countries with higher wages, for example, in service sector, may employ automation sooner than countries with lower wages. India as a developing country still suffers from the lack of skilled manpower in the automation sector.

### Large investments.

Automation requires a significant amount of capital spending as it involves installation of hardware such as robots with huge levels of dexterity, devices

with sensory perception competency and wheels that allow mobility. This leads to huge initial costs. However, costs decline over time, making automation more competitive with human labour.

Automation devices often require continuous modifications and maintenance in order to meet the ever-changing functionality of systems. As systems become more advanced, software updates are required. Developing and deploying these devices require huge investments in the years to come. Thus it is crucial to deploy a system that is flexible enough to evolve and adapt.

**Technology.** Automation can be deployed only in areas that are technologically advanced enough. Machines are required to reach a certain level of competency in order to carry out automated activities.

**Acceptance.** The rate of automation adoption is affected by regulatory approval and the response to such technologies by the public. Businesses and government policies may slow down adoption. In some cases,